BADDA SATHWIK
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About Me

I'm Sathwik Badda — a self-evolving system in human form, wired like a reinforcement learning agent: fast-adapting, always optimizing, and fueled by curiosity. I don't just learn; I evolve with every iteration — building smarter systems, sharper ideas, and scalable impact. If you're seeking someone who turns feedback into fuel and ambiguity into innovation — I'm already in your future.

Skills

- **Programming Languages:** Python, JavaScript
- **Artificial Intelligence:** NLP, Computer Vision, Reinforcement learning
- **Machine Learning:** PyTorch, Scikit-learn, TensorFlow, Keras
- **Deep Learning:** Neural Networks, CNN
- **Generative AI:** RAG, Lang Chain, Knowledge Graphs
- **Agentic AI:** AI agents, N8n, Multi Agentic Workflows, Lang graph
- **Data Visualization**: Matplotlib, Seaborn, Tableau, Cloud Computing
- **Databases**: MySQL, MongoDB, Firebase, Supa base, neo4j
- **DevOps**: Docker, GitHub Actions, Dvc, LangSmith, Lang Fuse,

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Experience

Gen AI Intern—*Diebold Nixdorf | Sep 2025 – Jan 2026*

- Developed an automated pipeline for Mandate Document Generation for VisaNet and MasterCard, streamlining the entire workflow from document ingestion to final output.
- Integrated Azure Document Intelligence to preprocess raw mandate forms—extracting text, detecting structure, cleaning noisy fields, and converting unstructured PDFs into machine-readable, markdown-based structured data.
- Implemented validation logic where extracted content is cross-checked using LLM-based reasoning, ensuring accuracy, completeness, and compliance with VisaNet and MasterCard standards.
- Designed an end-to-end automated workflow that processes documents, generates standardized outputs, and exports final mandate files in preferred formats (PDF/Docx), reducing manual effort by 95%**.**
- Added custom business rules and prompt-engineering layer to make mandate generation consistent, audit-ready, and regulator-compliant**.**
- Developed a scalable multimodal RAG pipeline that integrates text and image data using LMDB-based image retrieval with placeholder mapping.
- Implemented recursive parent–child chunking and FAISS indexing to improve retrieval precision and contextual accuracy.
- Engineered a cost-efficient system delivering sub-second multimodal responses without relying on large multimodal LLMs.

AI Engineer Intern—*Relai | May 2025 – August 2025*

- Developed AI-Powered Chatbots: Utilized Lang Chain and Lang Graph to design and implement conversational agents for social media analytics and customer support, integrating with platforms like WhatsApp to enhance user engagement.
- Integrated Meta APIs: Worked with Meta's Graph API and WhatsApp Cloud API to handle real estate inquiries, deliver investment recommendations, and enable real-time, personalized chatbot interactions
- Leveraged LLMs & Memory: Integrated OpenAI API and Claude API as LLM backends for intelligent response generation. Used Lang Chain Buffer Memory storage to preserve context and multi-turn conversation history.
- Built Full-Stack Web Application: Contributed to the development of a real estate platform using React.js, TypeScript, and Supa base, allowing users to explore listings, invest in properties, and receive AI-powered suggestions.
- Implemented AI-Powered Call Bot Integration: Developed an Exotel-based AI call bot connected to WhatsApp and integrated with Zoho CRM to automate voice interactions, capture leads, and sync customer data for improved sales workflows

- Built Multi-Source Sentiment Analysis Pipeline: Designed a data scraping and NLP pipeline to collect location-specific content from news articles, YouTube, Reddit, and Twitter, performing real-time sentiment analysis to generate actionable insights for property investment decisions.

AIML Intern— *Ediglobe | June 2024 – August 2024*

• Developed Predictive Analytics and Recommendation Models: Collaborated with the data science team to build machine learning models aimed at forecasting user behavior and providing personalized recommendations.
• Data Preprocessing and Feature Engineering: Conducted extensive data cleaning and feature extraction processes to prepare datasets for model training, ensuring high-quality input data.
• Hyperparameter Tuning and Model Optimization: Engaged in fine-tuning model parameters to enhance performance metrics, employing techniques such as grid search and cross-validation.
• Pipeline Deployment and Documentation: Assisted in deploying machine learning pipelines for internal testing environments and prepared comprehensive documentation for client presentations and future reference**.**

Projects

- **Automating Manim Library**
  Developed an agentic workflow to generate 3Blue1Brown-style educational animations from user prompts using a 5-agents LangChain, Lan graph architecture with an orchestrator. Integrated Claude for script + Manim code generation with automated debugging and rendering. Added GTTS narration to produce complete end-to-end explanatory videos.
- **AI Agentic Platform:** Built a no-code agentic AI automation platform(**Like N8n)** that enables users to visually create and execute LLM-powered workflows using drag-and-drop nodes, tools, and retrieval systems.
- **AI Website Maker:**
  Developed an AI-powered web application utilizing Gemini 2.5 Pro LLM and integrated agents to assist users in website development, offering real-time code error resolution by referencing platforms like GitHub, and providing live code previews through Web Container.
- **AI-Powered Assistive Vision System for the Visually Impaired**:
  Developed a multi-modal AI platform using LLaVA, YOLOv8, and OCR for real-time scene narration, object detection, and text reading. Integrated Lang Chain agents for summarization, translation, and guidance, optimized for edge and cloud deployment.

Research and Achievement's

- **Solar-Powered, Farmer-friendly Sugarcane Cutter:**
  Patent application ID: ***202541080519***
  Title- ***Hybrid-Powered Sugarcane Harvesting Device***
  Date of Filing - ***25/08/2025***
  Status: ***Published***
  Designed and developed a cost-effective, eco-friendly sugarcane cutter that ensures minimal crop damage and operates on a hybrid power model.
- **Hackathon Participation:**
  1. Participated in **HackSavvy24-1(MGIT-2024).** Collaborated with a team to develop an innovative solution for Deep Fake Detection
  2. Participated in **ISRO Bhartiya Antariksh Hackathon (2025).** Collaborated with a team to develop a Rag pipeline for creating a chatbot of mosdac (a satellite website) scraping the entire data

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Education

B.Tech in Artificial Intelligence and Machine Learning
Woxsen University, Hyderabad | August 2022 – Expected August 2026

- *Current CGPA: 3.3/4*